Simple Rules, Complex Behavior KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

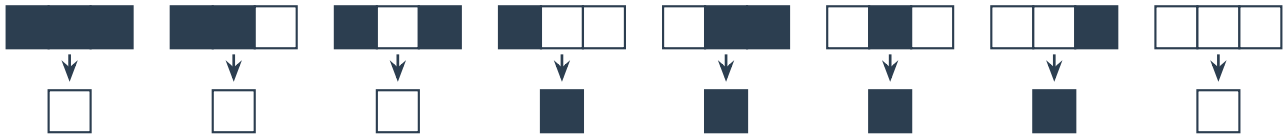
1. The wave. Was anyone in charge of the wave? no

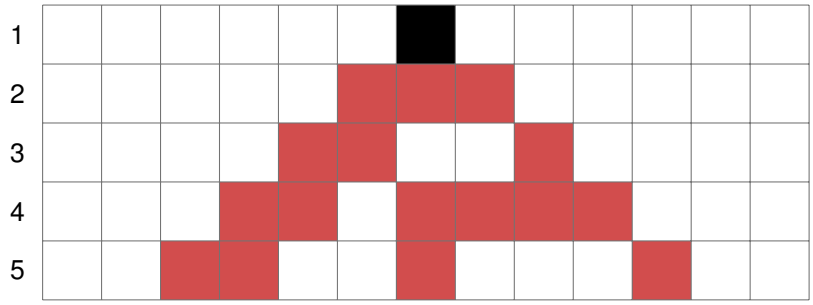
Did the person you watched move down the row? no So what moved? the wave (the pattern)

When many simple parts make something bigger that nobody planned, it's called emergence.

2. Make a few rows. Look at a square and its two neighbors in the row above. Find the same 3 colors on the card. Color the square below. (Outside the grid, everything is white.)

Rule 30: what the middle cell becomes next





Do your rows match the tool's first rows? yes

3. Explore the rules. Draw each rule. Circle what it looks like.

Rule	What does it look like?			
90	never changes	a pattern that repeats	chaos	complex (repeats)
110	never changes	a pattern that repeats	chaos	complex (complex)
4	never changes	a pattern that repeats	chaos	complex (never changes)
30	never changes	a pattern that repeats	chaos	complex (chaos)

My own rule. My rule number is Answers vary. It looks like _____

4. Watch the Game of Life.

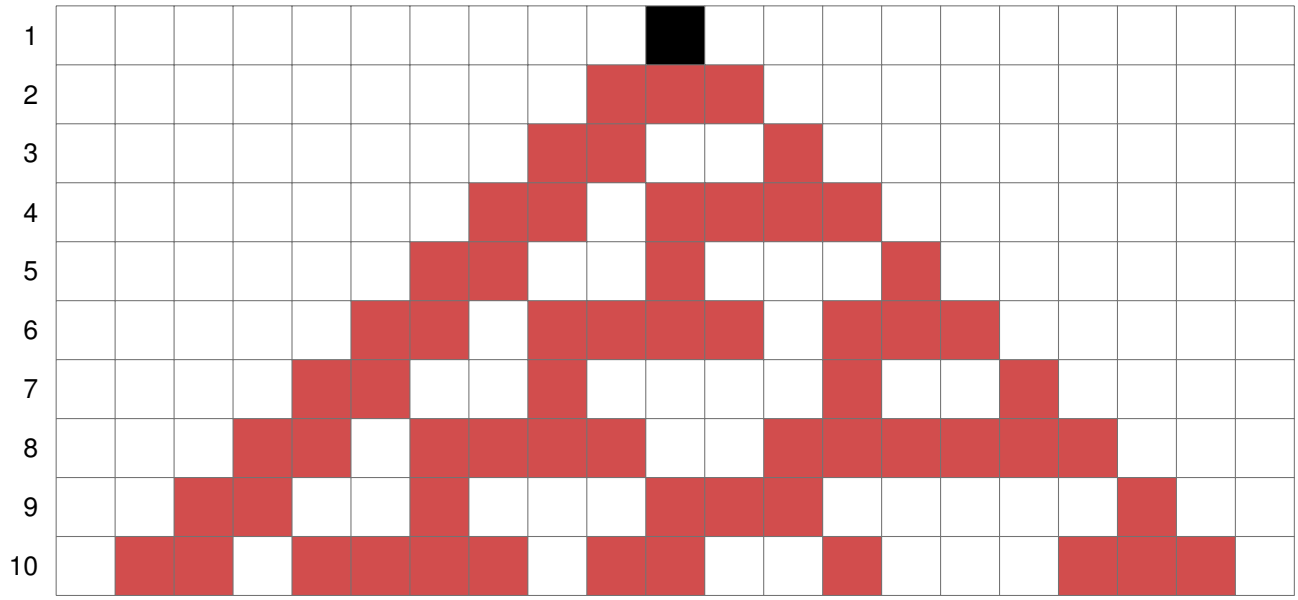
What did the glider do? It moved across the screen (at an angle).

Did any single cell move? no So what moved? the pattern

5. Talk about it. Can simple parts make something smart?

I think simple rules **can** / **cannot** make something smart, because Answers vary.

Go Further (Day 2). Be Rule 30 as a class. Color each new row the class shows.



Game of Life. A black cell with 2 or 3 black neighbors stays black. A white cell with exactly 3 black neighbors turns black.